

WEST POINT PERTINENT DATA

GENERAL

Location (damsite) – Troup County GA, Chambers County AL, miles above mouth of Chattahoochee River	201.4
Drainage area, Buford Dam to West Point Dam, square miles	2,406
Total drainage area above West Point Dam, square miles	3,440

RESERVOIR

Top of conservation pool elevation, summer/winter - feet NGVD29	635/628
Bottom of conservation pool, feet NGVD29	620.0
Top of flood risk management pool - feet NGVD29	641
Peak pool for spillway design flood – feet NGVD29	646.4
Peak pool for standard project flood – feet NGVD29	640.9
Total flood risk management storage, winter (elev. 641 - 628) - acre-feet	332,503
Total flood risk management storage, summer - (elev. 641 - 635) - acre-feet	170,271
Conservation storage (elevation 635 to 620) - acre-feet	306,131
Inactive storage below elevation 620 feet NGVD29 - acre-feet	298,396
Area at top of conservation pool (elev. 635) – acres	25,864
Shoreline at maximum power pool (elev. 635), miles	604

TAILWATER ELEVATION

Three units running at max capacity (18,000 cfs) – feet NGVD29	565.5
Small unit running (675 cfs) – feet NGVD29	557.8
Minimum (no flow) – feet NGVD29	557.0

DAM

Length, concrete sections/earth embankments - feet	896/6,354
Top of dam elevation – feet NGVD29	652.0

SPILLWAY

Type	Concrete gravity ogee
Gross length – feet	350
Tainter gates, number	6
Tainter gates, width x height – feet	50 x 41

POWER PLANT

Generating capacity (declared*) MW (2 units @ 42, 1 unit @ 3)	87
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* Declared generating capacity is defined as the plant's operational capacity declared on a weekly basis to the power marketing agency. The value may vary slightly from week to week depending on factors such as head and cooling capabilities; values shown are the nominal values reported.

STREAM FLOW AT DAMSITE - Before filling lake*

Period of record 1896–1973 (Before construction of West Point Dam
(River flow was first interfered with in May 1967 when the river was diverted
through a channel cut in the east bank.)

Average flow cfs	5,628
Minimum monthly flow cfs (Sept. 1925)	333
Maximum monthly flow cfs (March 1929)	32,700

Period of record 1974–2013, (after construction of West Point Dam)

Average flow cfs	4,712
Minimum monthly flow, cfs (Aug. 2002)	823
Maximum monthly flow, cfs (March 1990)	18,835

Estimated bankfull discharge, cfs	40,000
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RESERVOIR

Top of conservation pool, feet NGVD29 – summer level	635.0
Top of conservation pool, feet NGVD29 – winter level	628.0
Minimum conservation pool, feet NGVD29	620.0
Elevation of spillway design flood	646.4
Elevation of standard project flood	640.9

Storage volumes, acre-feet

Total storage at maximum flood pool (elev. 641)	774,798
Total storage at top of conservation pool (elev. 635)	604,527
Permanent flood risk management storage – (elev. 635 to elev. 628)	162,232
Surcharge flood risk management storage storage (elev. 641 – elev. 635)	170,271
Conservation storage – summer (elev. 635 to elev. 620)	306,131
Conservation storage – winter (elev. 628 to elev. 620)	143,899
Inactive storage below minimum power pool, elev. 620	298,396

Reservoir area, acres

Maximum flood pool (elev. 641)	31,993
Top of conservation pool, summer level	25,864
Top of conservation pool, winter level	20,615
Minimum conservation pool,	15,512

Shoreline at maximum power pool (elev. 635), miles

Mainland shoreline	604
Islands shoreline	14

DAM

Concrete gravity and earth embankment

Length, feet	
Concrete sections	896
Earth embankments	6,354±
Elevations, feet, NGVD29	
Top of dam	652.0
Streambed	556
Spillway Crest	597.0
Top of tainter gates, closed	638.0
Service bridge	652
Maximum height, feet	
Concrete section	132±
Earth embankment	78
Maximum top width, feet	
Concrete (non-overflow)	20.5
Roadway	15.0
Earth embankment	25.0
Sideslopes, earth embankment	
Upstream	1 on 3
Downstream	1 on 2.5
Spillway	
Type	Concrete gravity ogee
Gross length, feet	350
Clear opening, feet	300
Piers, number	5
Piers, width, feet	10
Tainter gates, number	6
Tainter gates, width, feet	50
Tainter gates, height, feet	41
Type-of bucket	Roller
Radius of bucket, feet	50
Bucket lip, elevation	550.64
Intake-powerhouse section	
Length, feet (estimated)	321
Nonoverflow section	
Length, feet	185

POWER PLANT STRUCTURE

Right bank of the river between the earth dike and the spillway

Size of building, feet	
Length	321
Width	164.5
Elevations, feet, NGVD29	
Bottom of substructure (main units)	504.5
Low point of draft tube	
Main units	509.0
Small unit	538.0
Centerline of distributor	
Main units	561.0
Small unit	556.0
Turbine room floor	
Main units	582.0
Small unit	567.0
Elevations, feet NGVD29	
Control room	582.0
Erection room	602.0
Transformer deck	602.0
Draft tube deck	599.0
Intake deck	652.0
Crane runway rail	626.0
Roof (high point of concrete slab)	642.08)
Top of parapet	645.25
Trash racks	
Number of openings,	
Main units	9
Small unit	1
Size of rack opening, each main unit, feet	17.0 x 44.0
Spacing of bars, center-to-center, inches	6
Intake gates – Vertical lift, single leaf, tractor	
Number, each	
Main units	3
Total of	6
Small unit	1
Size (clear opening), feet	
Main units	17.0 x 31.06
Small unit	9.0 x 9.0
Method of handling	Individual hydraulic hoists

Draft tube gates	
Type	Double leaf, vertical lift, slide
Number	
Main units (each)	3
Small unit	1
Number provided	3
Size (clear opening), feet	17.67 x 25.0
Method of handling	Gantry Crane

MAIN GENERATING UNITS

Number, initial	2
Number ultimate	3
Speed, rpm	100
Spacing of units, center-to-center, feet	77
Turbines	Fixed blade propeller, counter-clockwise rotation
Capacity, hp each	
at 52.5 feet net head,	45,000
at 58.0 feet net head	51,500
at best gate	48,000
Spiral cases	Concrete semi-spiral, three discharge passages
Draft tubes	Concrete elbow, three discharge passages
Generators	Vertical shaft, enclosed, self-ventilated, with thrust and guide bearing below rotor
Rating, each, Normal, continuous, kw	48,840
Power factor	0.90
Voltage (60 Hertz, 3-phase)	13,800

SMALL GENERATING UNIT

Speed, rpm	327
Location	In erection bay block
Turbine	Fixed blade propeller, counter-clockwise rotation
Capacity, hp	
at 25 52.5 feet net head	3,500
at 77.feet net head (rated)	5,400
Number	1
Spiral case	steel, one discharge passage
Draft tube	concrete elbow, one discharge passage
Generator	Vertical shaft, enclosed, self-ventilated, with thrust and guide bearing above rotor, and guide bearing below the rotor
Rating, kw	
Normal, continuous	3,375
Normal, continuous, 60 °C rise	3,750
Maximum, continuous capability	4,312
Power factor	0.90
Voltage (60 Hertz, 3-phase)	4,160
Average annual energy (1976–2013), MWH	184,106

TAILWATER ELEVATION, FEET, NGVD29

Spillway design flood (423,900 cfs)	604.0
Standard project flood (249,500 cfs)	595.0
Maximum flood of record (73,200 cfs)	579.0
Three units running at max capacity (18,000 cfs)	565.5
Small unit running (675 cfs)	557.8
Minimum (no flow)	557.0

HEAD, NET FEET

Maximum	
Small unit (675 cfs)	77.0
One main unit (6,300 cfs)	73.0
Minimum	
Small unit only (675 cfs)	62.0
Both main units and small unit (18,000 cfs)	52.5
Mean operating	
Small unit (675 cfs)	62.0
Both main units (16,825 cfs)	63.2
Rated	
Small unit	77.0
Main units	58.0